

NuraVolt

5-Step PV Data Quality Validation Checklist

Quick validation for sensor calibration, timestamp sync & outlier detection

Purpose: Use this checklist monthly (minimum) or after any SCADA/sensor maintenance to validate irradiance data quality. Poor data quality causes 14% yield loss attribution errors and 8-15% false alarms. Print this checklist and keep results on file for warranty claims and performance guarantees.

1 Sensor Calibration Check



Compare GHI sensor to clear-sky model

On clear day, measured GHI should be within $\pm 10\%$ of PVLIB clear-sky calculation for your location

✓ **Pass:** $<10\%$ deviation | ✗ **Fail:** $>10\%$ deviation



Check sensor-to-sensor agreement

If multiple sensors: all sensors should read within 5% of each other under uniform conditions

✓ **Pass:** $<5\%$ spread | ✗ **Fail:** $>5\%$ spread (investigate outlier)



Verify last factory calibration date

Sensor should have factory recalibration <2 years old (annually preferred)

✓ **Pass:** <24 months | ⚠ **Warning:** >24 months (schedule recalibration)

Tools: PVLIB Python library for clear-sky modeling | Reference sensor for cross-check | Calibration certificates

2 Timestamp Synchronization



Verify NTP server configuration

Confirm all devices (SCADA, weather station, inverters) use same NTP time source

✓ Pass: All synced to same NTP | ✗ Fail: Mixed time sources or manual time



Check SCADA-to-sensor timestamp alignment

Peak GHI timestamp should match peak inverter production timestamp within ± 10 seconds

✓ Pass: <10 sec offset | ✗ Fail: >10 sec (causes 14% yield loss attribution error)



Test clock drift over 24 hours

Monitor time offset between SCADA and sensor logger over 24-hour period

✓ Pass: <5 sec drift | ⚠ Warning: >5 sec/day (configure NTP sync more frequently)

Tools: NTP query (ntpq -p command) | SCADA timestamp logs | Cross-correlation of GHI vs production peaks

3 Outlier & Physical Limit Detection



Check for physically impossible values

GHI should never exceed 1,500 W/m² (extraterrestrial limit ~1,367 W/m² plus atmospheric effects)

✓ Pass: All values <1,500 | ✗ Fail: Values >1,500 (sensor error or data corruption)



Identify negative or zero values during daylight

GHI should be >0 when sun elevation >10° (no nighttime zeros during day)

✓ Pass: No daytime zeros | ✗ Fail: Zeros present (sensor or logger issue)



Check for "stuck" sensor readings

Sensor should vary continuously during day; constant value >15 minutes indicates sensor freeze

✓ Pass: No stuck values | ✗ Fail: Repeated identical readings >15 min

Tools: Automated outlier detection script | PVLIB for sun position | Data visualization (time series plots)

4 Missing Data & Gap Analysis



Calculate data completeness percentage

Irradiance data should be >98% complete over last 30 days (daylight hours only)

✓ **Pass: >98% complete** | ⚠ **Warning: <98% (investigate communication or logger issues)**



Identify gaps >15 minutes

Gaps >15 minutes require special handling (no interpolation; use clear-sky model or flag as invalid)

✓ **Pass: All gaps <15 min** | ⚠ **Warning: Gaps >15 min detected (document and flag)**



Review gap-filling protocol

Confirm SCADA uses acceptable interpolation (linear for <5 min gaps, clear-sky for 5-15 min, flag for >15 min)

✓ **Pass: Protocol documented and followed** | ✗ **Fail: Gaps zero-filled or inappropriately averaged**

Tools: SCADA data export with timestamp analysis | Python pandas for gap detection | PVLIB for clear-sky gap filling

5 GHI-POA Consistency Validation



Cross-check GHI vs POA sensor readings

POA (plane-of-array) should equal $GHI \times \cos(\text{angle of incidence}) \pm \text{diffuse corrections}$ (typically within 15%)

✓ **Pass: <15% deviation** | ✗ **Fail: >15% (sensor placement or soiling issue)**



Compare GHI-based expected production to actual

Calculate expected production from GHI + system specs; compare to actual inverter output (should match within 10%)

✓ **Pass: <10% PR deviation** | ⚠ **Warning: >10% (investigate soiling, shading, or equipment issues)**



Verify sensor soiling and cleaning schedule

Sensors should be cleaned weekly (more frequently than panels); check last cleaning date and visual inspection

✓ **Pass: Cleaned <7 days ago** | ⚠ **Warning: >7 days (sensor reads low, false soiling indication)**

Tools: PVLIB for GHI→POA transposition | PVsyst or SAM for expected production | Maintenance logs for cleaning dates

Need Help Implementing These Checks?

Get a free 15-minute data quality audit from our UAE-based team

[Schedule Free Consultation →](#)

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